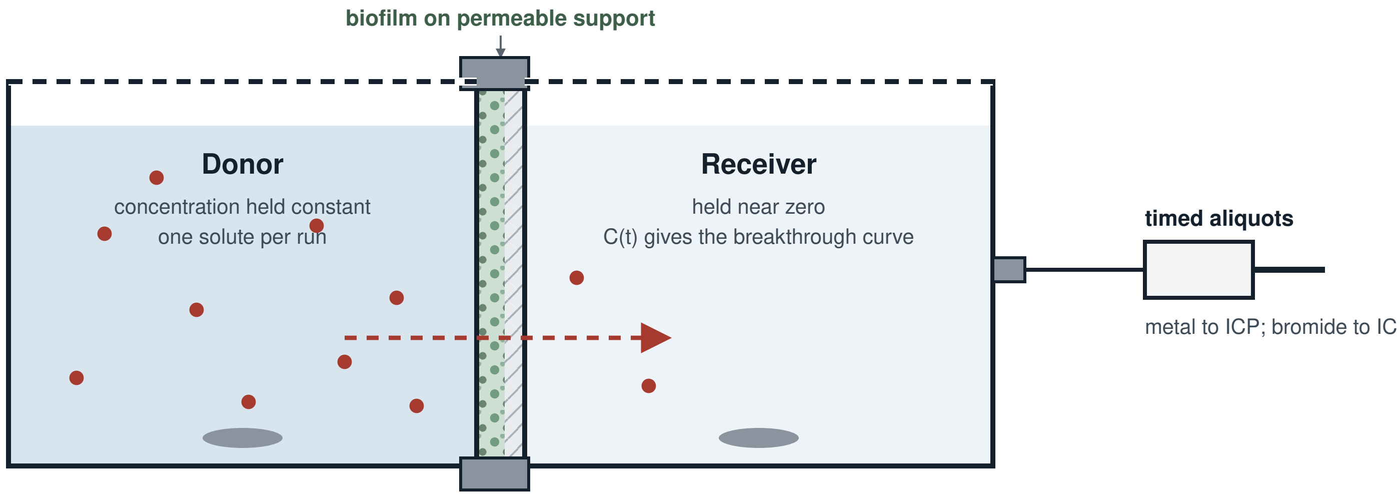
Phase 2 — two-chamber diffusion cell

Biofilm grown on a permeable support; transport read as receiver breakthrough



Both chambers stirred — unstirred boundary layers otherwise read as film resistance.

Paired blank: bare support

Identical cell and support, no biofilm, run alongside every measurement. Gives the support's own resistance and its wall sorption. Without it the result is film plus support, not film.

Edge seal checked separately: a leak reads as fast transport.

Open choice: the tracer run

Bromide is the tracer because $R \approx 1$: it is not retarded, so its breakthrough gives D_eff directly — the number the solver takes. If bromide is excluded from part of the pore space, that D_eff is not the metal's. Whether an EPS gel does that is untested. At Hanford 300A tritiated water resolved pore volume bromide did not (Hay et al. 2011, doi:10.1029/2010WR010303). Tritiated water's approval burden: a question for Dr. Deng.

What the metal run yields

Breakthrough lag gives a lumped transport parameter, not a diffusivity alone: sorption retards it, so the metal run yields D_app. Separating the two needs the partition coefficient from the Phase 1 suspended isotherm.

That ordering is why Phase 1 comes first.

What this number is not

The film-scale effective diffusivity measured here is not the reactor-scale dispersion coefficient that currently carries the same name in radial_parms ($1\times10^{-3}\ \mathrm{cm^2\ s^{-1}}$, about 43× water's self-diffusi
It is the quantity slab_parms carries as D_eff ($1\times10^{-5}\ \mathrm{cm^2\ s^{-1}}$) — the one default in that file below water self-diffusivity.